

SECOND SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

TEXTILE FIBRES
MODEL QUESTION PAPER

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all questions in one word or one sentence. Each question carries one mark.

(9 x 1 = 9 Marks)

1	The empty space at the Centre of the Cotton fibre is called	M1.02	U
2	Percentage of Cellulose in raw Cotton is	M1.02	U
3	The bast fibre used for currency making is	M2.01	U
4	The fibre forming protein of wool is	M2.04	U
5	The chemical reaction in which two or more molecules combine to form larger molecules that contain repeating structural units is called	M4.01	U
6	An example of modified cellulosic rayon is	M3.02	U
7	The acid used in the coagulating bath of Viscose rayon spinning is	M3.01	U
8	An example of manmade polyamide fibre is	M4.03	U
9	Hexa methylenediamine is used for making ----- fibre.	M4.03	U

PART B

II. Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24 Marks)

1	Explain the botanical classification of cotton.	M1.02	U
2	Draw the microscopic cross-sectional and longitudinal appearance of wool.	M2.04	U
3	Discuss the uses of Viscose Rayon.	M3.01	U
4	Write a short note on Modal fibres.	M3.03	U

5	Explain the chemical properties of Acetate Rayon.	M3.02	U
6	Discuss the properties of Polynosic Rayons.	M3.01	U
7	What do you mean by Linear and branched polymers?	M4.01	U
8	Write a short note on Elastomers.	M4.01	U
9	State the terms Monomer and Polymer.	M4.01	U
10	Discuss the uses of Nylon 6.6	M4.03	U

PART C

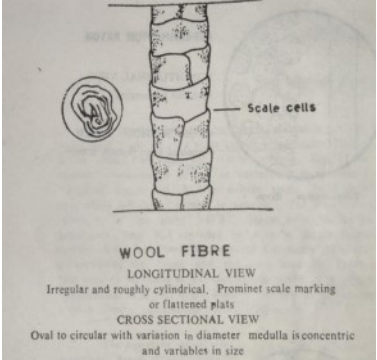
Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

III	Explain the types of commercial cotton	M1.02	U
	OR		
IV	Explain the essential properties of textile fibre.	M1.01	U
V	Explain the extraction of linen from the Flax plant	M2.01	U
	OR		
VI	Explain the structure of wool with a sketch.	M2.04	U
VII	Discuss the various steps involved in the manufacture of Acetate Rayon.	M3.02	U
	OR		
VIII	Explain the various steps involved in the manufacture of Viscose Rayon.	M3.01	U
IX	Explain the manufacture of Nylon 6 6 from Raw materials with a flow chart.	M4.03	U
	OR		
X	Describe the principle of Melt spinning with a sketch.	M4.02	U
XI	Explain the classification of textile fibres on the basis of chemical composition.	M1.01	U
	OR		

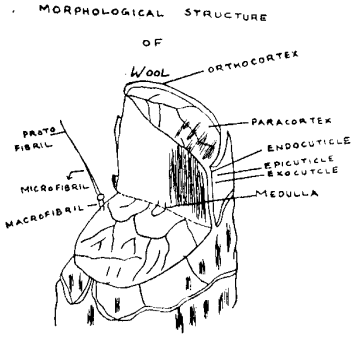
XII	Explain the Structure of cotton fibre with a sketch.	M1.02	U
XIII	Describe the physical and chemical properties of Jute.	M2.02	U
	OR		
XIV	Explain Various stages involved in the production of Raw Silk from cocoons.	M2.03	U

Scoring Indicators - TEXTILE FIBRES

q no	scoring indicators	split score	total score
I	PART A		
1	Lumen		
2	85.5 %		
3	Linen		
4	Keratin		
5	Polymerization	9	9
6	Acetate Rayon		
7	Sulphuric acid		
8	Nylon		
9	Nylon 6.6		
II	PART - B		
1	The cotton plant belongs to the natural order of Gossypium Mallow family. There are mainly four species. 1. Gossypium herbeceum 2. Gossypium hirsutum 3. Gossypium peruvianum 4. Gossypium barbadense . The first two (Gossypium herbeceum , Gossypium hirsutum)are cultivated in America and India .The third as its name implies originated in Peru and spread to other South American countries. Gossypium barbadense produces fine silky Sea Island cotton and is cultivated in Gulf of Florida.	3	
2	 WOOL FIBRE LONGITUDINAL VIEW Irregular and roughly cylindrical. Prominent scale marking or flattened plates CROSS SECTIONAL VIEW Oval to circular with variation in diameter medulla is concentric and variables in size	3	

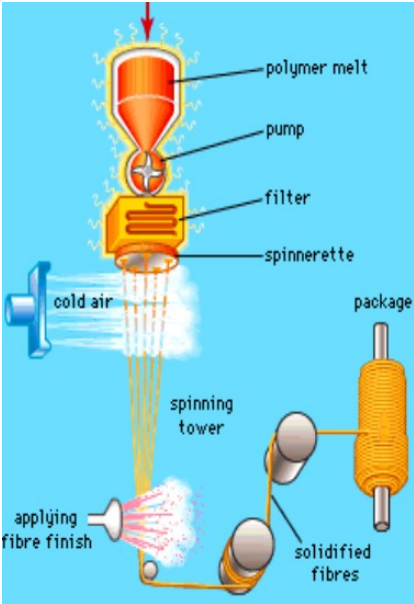
3	It is used practically in all types of apparel fabric. Ladies dress material such as georgettes chiffons, satins, crepes, velvets etc. are commonly made from Viscose rayon filament yarn. Home furnishings such as Curtains, bedspreads, cushions quilt covers, table cloth and upholstery material are also made from Viscose filaments. Viscose filament yarn is also used in knitted fabrics to some extent. Viscose staple is widely used in the production of nonwoven fabrics goods such as surgical, cosmetic and baby care products, hospital gowns, sanitary napkins, pillow cases, mattress cover etc.	3	
4	Modal is a type of rayon, a semi-synthetic cellulose fibre made by spinning reconstituted cellulose. Modal is used alone or with other fibres (often cotton or spandex) in clothing and household items like underwear, bathrobes, towels, and bedsheets. Modal is processed under different conditions to produce a fibre that is stronger and more stable when it is wet than standard rayon. Modal has a soft feel, similar to cotton. The fabric has been known to pill less than cotton due to fibre properties and lower surface friction.	3	
5	Effect of acids : Decomposed by strong acids; dissolved by acetic acid Effect of Alkalies : good resistance to weak alkalies, strong alkali destroys Resistance to Mildew - not affected by mildew	3	
6	Sunlight - weakened by sunlight Strength is higher, elongation at break lower, the ratio of wet strength to dry strength is much higher, there is less swelling in water, the Degree of Polymerisation is higher and there is microfibrillar structure in the polynosic that is never found in standard viscose.	3	
7	The simplest polymer is a linear polymer. A linear polymer is simply a chain in which all of the carbon-carbon bonds exist in a single straight line. Branched polymers occur when groups of units branch off from the long polymer chain. These branches are known as side chains and can also be very long groups of repeating structures.	3	

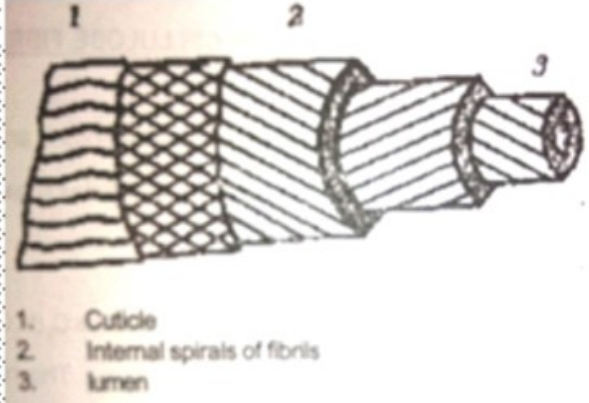
9	Elastomeric fibers are those fibers that possess extremely high elongations and that recover fully and rapidly from high elongations up to their breaking point. Elastomeric fibers are made up of molecular chain networks that contain highly amorphous areas joined together by crosslinks. On elongation, these amorphous areas become more oriented and more crystalline in nature. elongation continues until the crosslinks in the structure limit further molecular movement.	3																													
10	A monomer is a single atom, small molecule, or molecular fragment that, when bonded together with identical and similar types of monomers, form a larger, macromolecule known as a polymer. A polymer is a large molecule, or macromolecule, composed of small repeating singular molecular structural units called monomers. The repeating molecular units are joined together chemically through covalent bonds.	3																													
III	- Nylon is a high strength fibre. It is used for making fishing nets, ropes, parachutes and type cords. - It is used for making fabrics in textile industry. - Crinkled nylon fibres are used for making elastic hosiery	7																													
IV	PART - C <table border="1" data-bbox="329 1211 1166 1669"> <thead> <tr> <th>Name of Cotton</th><th>Length in inches</th><th>Fineness in inches</th><th>Count Spinnable</th></tr> </thead> <tbody> <tr> <td>Sea island</td><td>2 inches & more</td><td>1/ 1500 inches</td><td>200^s</td></tr> <tr> <td>Egyptian</td><td>1.5 to 1.75 inches</td><td>1/ 1500</td><td>70^s</td></tr> <tr> <td>South American</td><td>1 to 1.5 inches</td><td>1/ 1300</td><td>70^s</td></tr> <tr> <td>American</td><td>1 to 1.25 inches</td><td>1/1200</td><td>Coarser counts</td></tr> <tr> <td>Indian</td><td>0.6 to 0.8 inch</td><td>1/ 1200</td><td>Coarser counts</td></tr> <tr> <td>China</td><td>0.5 to 0.7 inches</td><td>1/ 1200</td><td>Coarser counts</td></tr> </tbody> </table>	Name of Cotton	Length in inches	Fineness in inches	Count Spinnable	Sea island	2 inches & more	1/ 1500 inches	200 ^s	Egyptian	1.5 to 1.75 inches	1/ 1500	70 ^s	South American	1 to 1.5 inches	1/ 1300	70 ^s	American	1 to 1.25 inches	1/1200	Coarser counts	Indian	0.6 to 0.8 inch	1/ 1200	Coarser counts	China	0.5 to 0.7 inches	1/ 1200	Coarser counts	7	
Name of Cotton	Length in inches	Fineness in inches	Count Spinnable																												
Sea island	2 inches & more	1/ 1500 inches	200 ^s																												
Egyptian	1.5 to 1.75 inches	1/ 1500	70 ^s																												
South American	1 to 1.5 inches	1/ 1300	70 ^s																												
American	1 to 1.25 inches	1/1200	Coarser counts																												
Indian	0.6 to 0.8 inch	1/ 1200	Coarser counts																												
China	0.5 to 0.7 inches	1/ 1200	Coarser counts																												
V	1. Staple Length 2. Tensile Strength 3. Fineness 4. Uniformity 5. Spinnability	7																													

VI	(Brief description) 1)Rippling The seeds and the leaves are removed from the stalks of flax plant by passing the stalks through a coarse comb. This process is called rippling. (2)Retting After rippling the stalks are retted , that is the outer bark is allowed to rot .This process when properly carried out, results in the breakdown of the bark and the release of the flax fibre without damage . (3) Breaking The stalks become partially separated from the fibre when the wet plants are laid in the field to dry .When the decomposed woody tissue is dry, it is crushed by being passed through fluted iron rollers. This breaking operation reduces the stalks to small pieces of bark called shives. (4) Scutching The Scutching machine removes the broken shives by means of rotating wooden paddles , thus finally releasing the flax fibre from the stalks. This operation can be done by hand as well as by machinery. (5) Hackling The simple combing process known as hackling, straightens the flax fibre , separates the short from the long staple and leaves longer fibre in parallel form. The wool is complex in structure and composed of three tissues , the cuticle , cortex and medulla.  Cuticle	7 5
-----------	--	-------------------

VII	The cuticle is formed of thin scales of hard and horny consistency. These overlap and protrude for about one third of their length , the ends being directed towards the tip of the fibre . The outermost layer of these scales is a tough membrane known as Epicuticle. Beneath this is situated the Exocuticle and the innermost layer is described as Endocuticle. Cortex About 90 percentage of the fibre is composed of cortical cells. It consists of countless, long, spindle shaped cells or cortical cells. The tensile strength, elastic properties and the natural colour of the wool are determined mainly by the nature of the cortical cells. Medulla Some of the coarser wool fibres have a hollow space in the centre called a medulla. The chemicals needed are: 1.Acetic acid 2. Acetic anhydride 3. Aceton 4.Sulphuric acid In addition to these, a good supply of water, free from iron is also needed. The stages in the manufacture of acetate rayon are as follows: (i) Cotton Purification : Cotton linters are first purified by kier boiling under pressure for 4-10 hours. They are washed, bleached with sodium " hypochlorite, washed and then dried. (ii)Pre-treatment : The purified cotton is steeped in glacial acetic acid to make it more reactive, so that will acetylate readily with acetic acid later. (iii) Acetylation : During acetylation, most of the hydroxyl (-OH) groups present in the cellulose are acetylated ie, replaced by acetyl group (CH₃COO). The actuated cellulose is treated with an excess of glacial acetic acid and acetic anhydride in a closed vessel fitted with a powerful stirrer. These are first thoroughly mixed together. When the mixing is completed sulphuric acid dissolved in 8 times its own weight of acetic acid is added. Now acetylation of cellulose takes place.At the end of about 8 hours all the cellulose will have been converted into cellulose	7	
		2	

	triacetate. The cellulose triacetate thus formed is known as primary acetate.		
VIII	(iv) Hydrolysis :In order to convert the primary acetate into acetone soluble secondary acetate, it is run together with excess of acetic acid and anhydride into water. (v) Preparation of Dope :The blended secondary acetate is mixed with 3 times Its weight of Acetone in a closed vessel. Acetone dissolves it slowly.After 24 hours it is completely dissolved. Dulling agent (Titanium dioxide) is added at this stage. Pigments may also be added if coloured fibres are required. The solution of cellulose acetate in acetone is called Dope. Dope is filtered. Deaerated and then run into a feed tank.	7	
IX	(vi) Spinning:The method of spinning of acetate rayon is known as dry spinning. The spinning solution called Dope is fed from the feed tank by pipes through a metering pump and a candle- filter to the spinnerets. The metering pump ensures a constant flow of Dope and the candle filter is used to give final filtration. This spinneret is located inside a spinning cabinet. The dope is extruded (allowed to pass) by pressure through the fine holes of the spinneret. In the cabinet hot air at a temperature of 100⁰C is circulated from the bottom. This evaporates all the acetone in the Dope.	7	
X	(i) Purification of cellulose: (ii) Conditioning of wood pulp: (iii) Steeping: (iv) Shredding: (v) Ageing: (vi) Churning: (vii) Dissolving: (viii) Blending, Filtration and Deaeration (ix) Ripening: (x) Spinning: (Brief description) The manufacturing process starts with the production of two chemicals hexamethylene diamine and adipic acid both of which contains 6 carbon atoms from coal. ▪ These chemicals are combined to form the nylon salt, which is dissolved in water and sent to the spinning mill. ▪ The nylon salt solution is made into a concentrated solution by heating in large evaporators. ▪ The concentrated solution is then heated in an autoclave	7	

	under the pressure and temperature. The polymerization takes place combining the two chemicals into polymers which are like a giant chains. The molten polymer is processed in a manner similar to nylon 6. The filaments obtained from spinning are stretched by drawing process. The filaments can be stretched from 2 to 7 times their original length. The molecules in the filament structure straighten out, become parallelized, and are brought very close together. MELT SPINNING In this process, a polymer is melted and heated to a suitable viscosity for fiber production. The melted polymer is pushed through a spinneret, which is a type of die consisting of several small holes. Each hole produces an individual fiber, and the number of holes on a spinneret defines the number of fibers in the yarn. The melted polymer fibers then passes through a cooling region and the fibers are combined to form a yarn and a spin finish is applied. The yarn is then drawn using several godets rolls with very good speed and temperature control to orient the molecules in the fibers and eliminate voids, making the yarn stronger. If the polymer is thermoplastic, then melt spinning should be used for higher productivity. Most of the manmade fibers, such as polyester, polypropylene, nylon and PGA, are produced using melt spinning..		
XI		7	
XII		7	

	Textile fibres <pre> graph TD A[Textile fibres] --> B[Man made] A --> C[Natural] B --> D[Synthetic] B --> E[Regenerated cellulose] D --> D1[Poly-acrylonitrile] D --> D2[Polyurethane] D --> D3[Poly-lactic] D --> D4[Polyamide] D --> D5[Polyester] E --> E1[Unmodified] E --> E2[Modified] E1 --> E1a[Lyocell] E1 --> E1b[Rayon] E2 --> E2a[Triacetate] E2 --> E2b[Acetate] C --> F[Polyamide Protein] C --> G[Cellulose] F --> F1[Silk] F --> F2[Wool] G --> G1[Cotton] G --> G2[Flax] G --> G3[Ramie] G --> G4[Jute] </pre>  Each hair consists of a single elongated cell. The upper end is tapered to a point while the lower end is open where it has been removed from the seed by a gin. The mature fibres contain many convolutions throughout its length . In a fully ripened cotton fibre , there are between 200 to 300 convolutions per inch. In case of dead fibres, the twist may be entirely absent , the cell walls are extremely thin and lumen is collapsed , so that the fibre look more like flat pieces of slightly twisted tape There is a primary wall of the young cell which becomes the outer skin of the mature fibres. The primary wall is covered with a circular layer of wax and pectic matter The secondary wall constitutes the bulk of the cellulosic component and is laid down in successive layers visible as rings in the cross	7
--	--	--------------------------------------

XIV	section of a swollen fibre. Within these rings, there are minute fibres deposit, fibres which are packed along side one another running in spirals. When cotton fibre is alive , the centre is filled with liquid nutrients and protoplasm , when the fibre dries the liquid disappears and an almost empty space known as the Lumen runs length wise through the fibre. Physical properties of Jute fibre : Length : The average length of flax fiber varies from 150-300 cm. Color: yellowish to brown Tensile strength: Stronger than cotton , Tenacity varies from 5 to 8 gm/denier. Fineness 6-20 microns Chemical properties of Jute fibre Effect of acids: Jute fiber is easily damaged by high concentrated acids under boiling condition. But it is not affected by low dense acids under cold condition if washed immediately. Effect of alkalis: Strength is degraded by strong alkalis. Affinity to Dyes: Less than cotton. Sunlight Effect: As lignin is present in jute fibers, it may be damaged by sunlight. Effect of microorganisms: Jute has resistance to microbiological attack Stifling Here the moth inside the cocoons are killed by steam or hot water. Grading or sorting To obtain uniform and good quality yarn , silk grading is done. Cocoons of different sizes are grouped together . Damaged as well as ill developed cocoons are rejected and used in the waste industry. Cooking	7	
-----	---	---	--

	The sorted cocoons are placed in 1 % hot soap solution at 90°C in a vat to soften the sericin gum and to enable the filaments to be drawn out easily as one continuous thread. Good cooking is essential for the success of subsequent operation and it is done for a few minutes. Brushing After cooking the unreelable, discontinuous and rough mass of fibres called floss is removed with a brush until the free end from which the continuous length is found. Floss is used in waste industry. Reeling The unwinding of the fine silk filaments from the cocoons is called reeling. Cocoons after brushing are transferred to a basin containing water kept at 60 °C . Reeling requires great skill as the operator must produce a uniform thread by combining the silk filaments in suitable fashion. the reeler must manipulate that the cocoons are so adjusted to produce uniform yarn.		
--	---	--	--